

QER – 6806

QER: GUMUSUT KAKAP PHASE 4 SF300505C004.11+12+13 - Zeta wire weld crack



Pipe section / Details

Item	Traceability / identification
Project number	300505C004
Project name	GUMUSUT KAKAP GERONGGONG JAGUS EAST PHASE 4
Pipe section	SF300505C004.11+12+13
ST number	ST127.70305
Affected layer / equipment	Pressure Vault Spiraling
Length affected	77.1m
Drawing number / MP code / STA / Dimension	Zeta 8 FI09 (MP234)

Overview of anomaly (Cont)

Problem Description:-

During the zeta spiraling process of GUMUSUT KAKAP GERONGGONG JAGUS EAST PHASE 4 SF300505C004.11+12+13 (ST127.70305), zeta wire weld crack was detected at length 77.1m on pipe.

To note that, the affected weld from winding process and it was detected during weld inspection after laid on the pipe.

Details of affected weld are as the following:-

- Weld number : 04-03
- Welding machine : BS13-11
- Bobbin number : 014

Root Cause:-

The root cause was suspected due to inlcusion as no anomalies were found on the machine (mechanically, power supply and forging distance).

Overview of anomaly

Problem Handling:-

The defective weld was rejected according to DP-QCP-08DAQT001. Therefore, un-spiraling was performed to remove the defective weld. In addition, based on DI-IND-MET1075, the subsequent welds shall be retrieved if reachable for the test and analysis. The subsequent welds were gone through the tensile test, and the result is detailed below.

Tensile results for the subsequent welds: -

- Weld 03-03 (N-1) = 586.7mPa
- Weld 06-03 (N+2) = 636.6mPa

The minimum criteria for the tensile test are 530mPa

To note that, N+1 weld are not reachable due to being laid on a pipe at length 59.9m.

Defect Photo

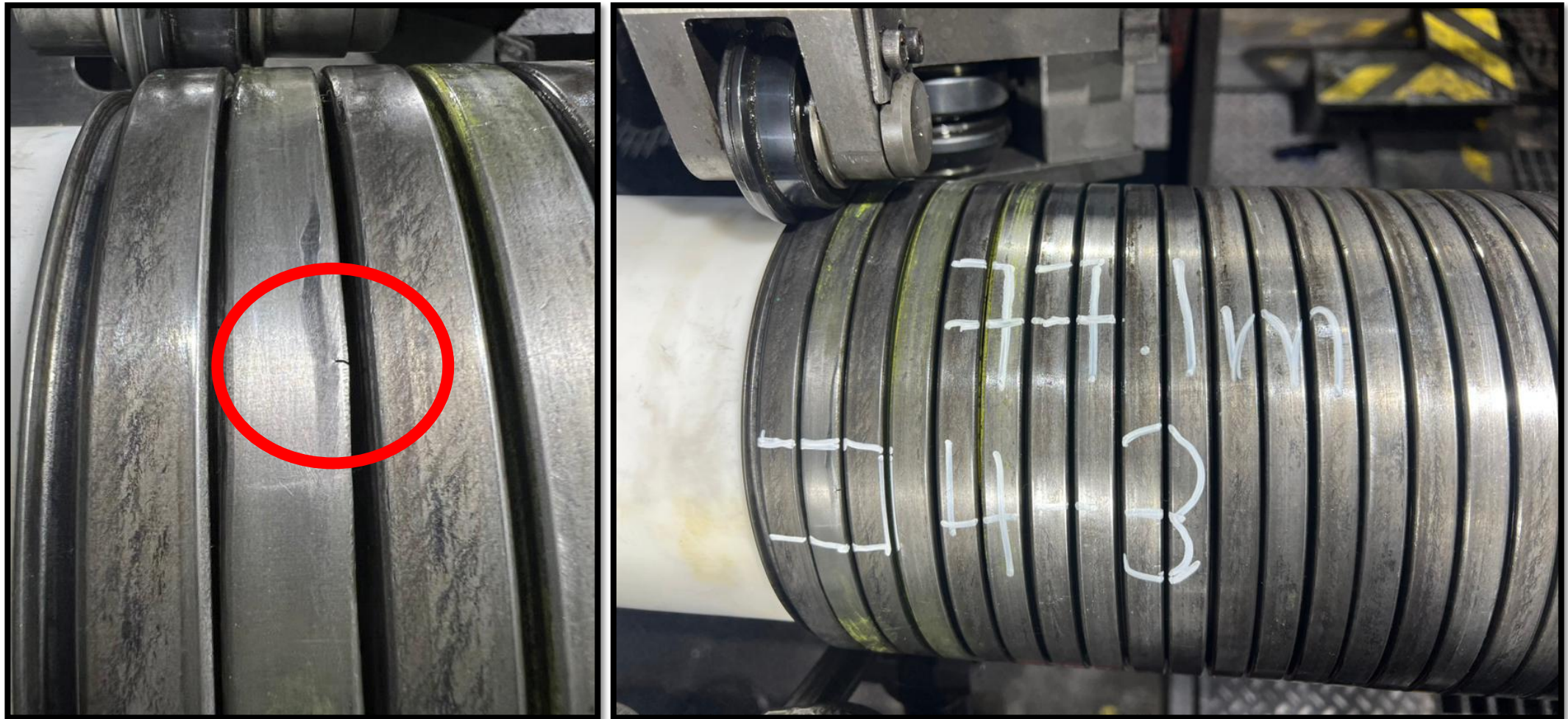


Figure 1 : Zeta wire weld crack was detected at length 77.1m on pipe.

Photo after repaired



Figure 2 : Un-spiraling was performed to remove the defective weld and replaced new weld (07-08)

THANK YOU

